



Paper



Code

Benchmarking Recurrent Event-Based Object Detection for Industrial Multi-Class Recognition on MTevent

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Motivation & Research Question

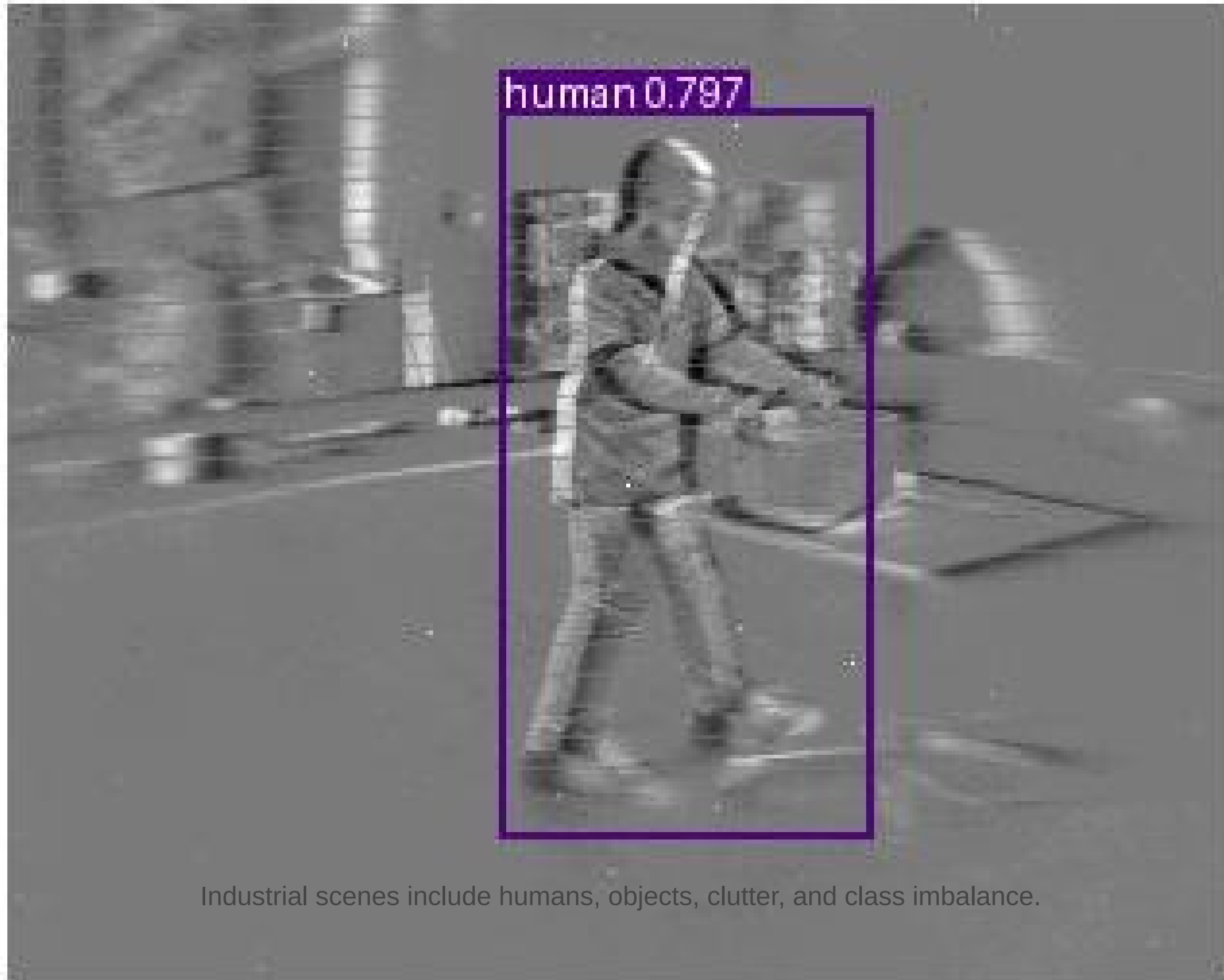
Why industrial event-based detection?

Event cameras offer high temporal resolution, high dynamic range, and reduced motion blur for fast robotic perception.
However, industrial multi-class recognition remains less explored than driving-focused event detection.

Research question

How much does recurrent temporal memory help, and how important is event-domain pretraining?

MTevent example



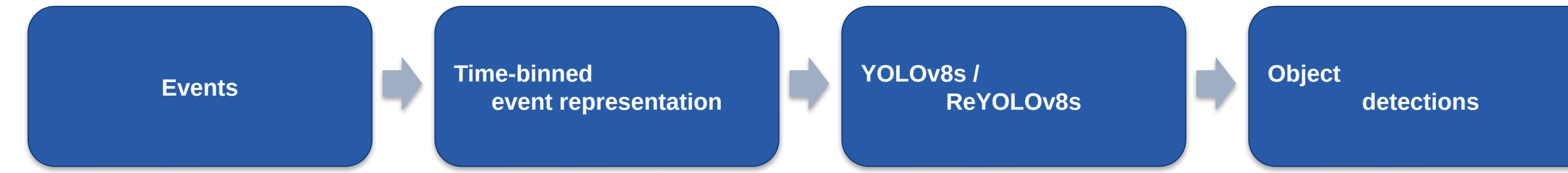
Dataset & contribution

- MTevent: industrial multi-class event dataset
- 17 object classes with strong imbalance
- Benchmark YOLOv8s vs. recurrent ReYOLOv8s

We study clip length, recurrent memory, and event-domain pretraining under the same validation metric.

Method & Experimental Setup

Detection pipeline



Events are accumulated into time-binned event frames, then evaluated with non-recurrent and recurrent detectors.

Model comparison

Baseline: YOLOv8s adapted for event-frame input; no recurrent temporal memory.
Recurrent model: ReYOLOv8s with ConvLSTM-based recurrent blocks and hidden states across event clips.

Experimental settings

Setting	Model	Initialization	Clip length
Baseline	YOLOv8s	Scratch	1
Recurrence	ReYOLOv8s	Scratch	3, 7, 11, 21
Pretraining	ReYOLOv8s	GEN1	3, 7, 11, 21
Pretraining	ReYOLOv8s	PEDRo	11, 21

Training/evaluation focus

- Clip length ablation: 3, 7, 11, 21
- Pretraining sources: GEN1 and PEDRo
- Metric: mAP50 on MTevent validation split

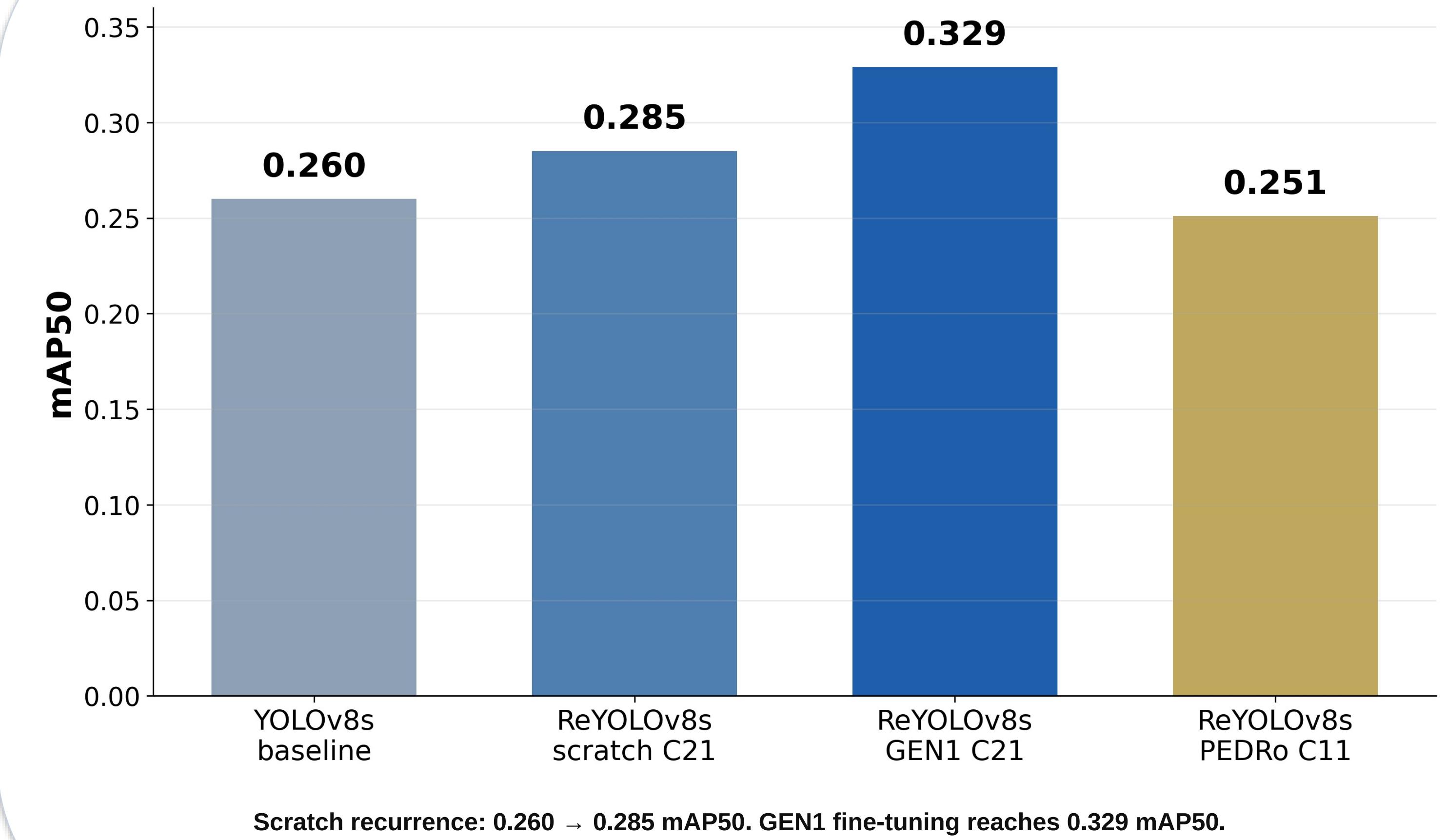
This design separates the effect of temporal memory from the effect of event-domain initialization.

Results & Takeaways

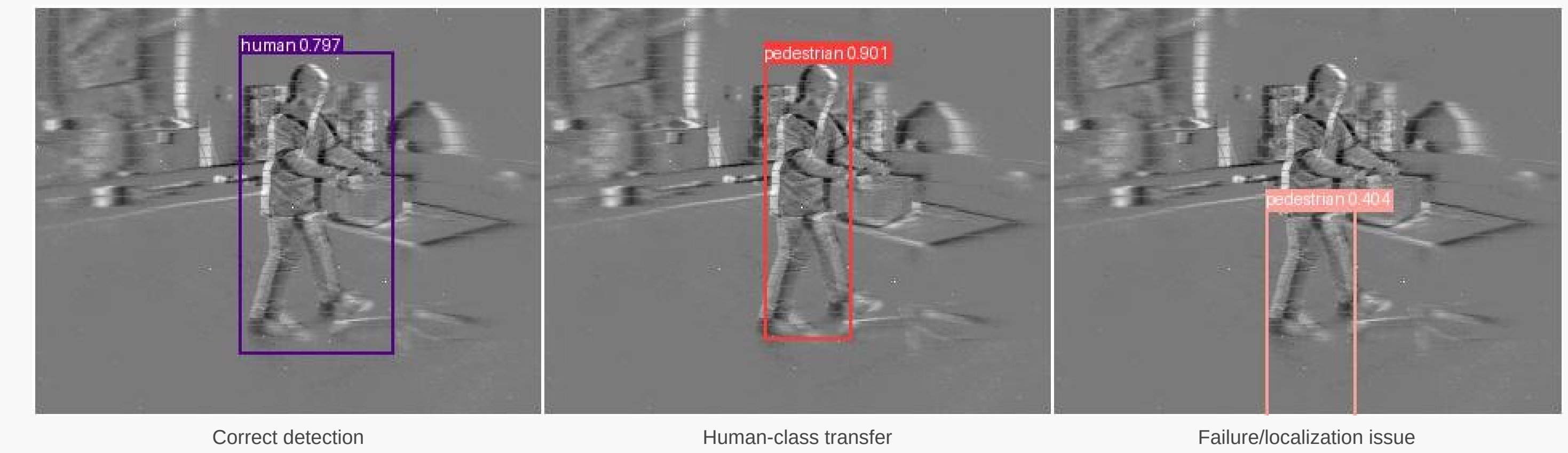
Main finding

Event-domain pretraining matters more than recurrence alone.

GEN1 pretraining gives the strongest improvement



Qualitative examples



Key takeaways

1. Recurrence gives a modest gain over the non-recurrent YOLOv8s baseline.
2. GEN1 event-domain pretraining gives the best performance.
3. Source-domain match is critical: PEDRo does not transfer well.
4. Remaining failures are linked to class imbalance and human-object interaction.